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Malicious URL Detection with Distributed Representation and Deep Learning

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Abstract. There exist numerous solutions to detect malicious URLs based on Natural Language Processing and machine learning technologies. However, there is a lack of comparative analysis among approaches using distributed representation and deep learning. To solve this problem, this paper performs a comparative study on phishing URL detection based on text embedding and deep learning algorithms. Specifically, character-level and word-level embedding were combined to learn the feature representations from the webpage URLs. In addition, three deep learning models, including Convolutional Neural Network (CNN), Bidirectional Gated Recurrent Unit (BiGRU), and Bidirectional Long Short-Term Memory (BiLSTM), were constructed for effective classification of phishing websites. Several experiments were conducted and various evaluation metrics were used to assess the performance of these deep learning models. The findings obtained from the experiments indicated that the combination of the character-level and word-level embedding approach produced better results than the individual text representation methods. Also, the CNN-based model outperformed the other two deep learning algorithms in terms of both detection accuracy and execution time.

Keywords. Malicious URL, phishing detection, distributed representation, deep learning

1. Introduction

Phishing is among various social engineering techniques that impersonate legal sources to trick the end-users into revealing their personal and private information [1]. Nowadays, phishers often use numerous attacking vectors, such as websites, emails, online social networks, etc., to conduct their phishing campaigns [2]. The common element in these attacking strategies is the presence of a malicious URL that will redirect the users to a

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compromised website [3]. According to the Phishing Activity Trend Report, in the fourth quarter of 2021, the number of phishing attacks has tripled from early 2020 [4]. In the same document produced by the Anti-Phishing Working Group (APWG), a total of 316,747 attacks were reported in December 2021, which was the highest monthly record in APWG's reporting history. These figures suggest that phishing is still currently on a rise, imposing a serious threat that needs to be addressed.

In an effort to mitigate phishing incidents, we propose a model for phishing detection using URL features extracted automatically from the websites via distributed representation and deep learning technologies. The proposed model consists of two tiers, with character-level embedding on the first tier and word-level embedding on the second tier. The outputs from these two tiers were combined and then fed into the deep learning classifiers to distinguish between malicious and benign URLs. We trained our proposed model on a dataset containing 73,575 URLs and assessed the model's performance through various evaluation metrics. We also conducted a comparative analysis on three different deep learning classifiers, such as CNN, BiGRU, and BiLSTM. The experimental results revealed that the CNN-based model achieved the highest detection accuracy of 99.10% in the shortest amount of time. Besides, the character-level and word-level embedding methods when combined in a hybrid approach produced better results than when being used separately. This paper continues our previous work [5,6] on phishing detection using deep learning technology. The outcomes of this paper are expected to assist researchers and security experts to select the best approach for developing and implementing their phishing detection solutions. Thereby, this paper makes the following contributions:

- Lexical features were extracted automatically from the website URLs to prevent manual feature engineering and third-party service dependency.
- Character-level and word-level embedding techniques were used in parallel to obtain the feature vectors of phishing websites simultaneously.
- A comparative analysis of phishing URL detection based on distributed representation and deep learning was conducted to provide a comprehensive comparison among state-of-the-art technologies.

2. Related Work

Natural Language Processing (NLP) is a text representation method to convert linguistic data into a numerical format that can be processed by machine learning models. Among numerous text representation techniques, distributed representation approach can capture the semantic and syntactic meaning of the input data and is the focus of this study. Keras is one of the most common distributed representation methods which is divided into two categories: character-level and word-level embedding. The former takes the input textual data as a sequence of characters, while the latter considers the input text as a sequence of words.

Character-level embedding technique has been used in deep learning-based phishing URL detection approaches. Srinivasan et al. [7] proposed a model called DeepURLDetect (DUD) using character-level embedding to encode the URLs and employing an ensemble deep learning architecture (CNN-LSTM) to classify them into malicious or benign websites. Results from the experiments showed that the proposed model performed better than other conventional text representation methods, such as

Bag-of-Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), or n-gram.

Besides character-level embedding, word-level embedding can also be used to learn the feature representation of website URLs. Wei et al. [8] suggested a lightweight solution to detect malicious URLs in a real-time and energy-saving manner. The author designed a phishing detection sensor that utilized word-level embedding configuration and CNN algorithm to improve the detection accuracy and reduce energy consumption. In comparison with other models using both character and word embedding, the proposed solution required less execution time and was more suitable for resource-constrained devices.

Realizing the individual strengths of character and word-level embedding techniques, several authors in the literature combined these two embedding methods in a hybrid model to effectively learn the feature representation and improve the overall performance accuracy. Rasheed et al. [9] investigated the robustness of phishing detection models by introducing several attacks and examining them against various CNN-based classifiers. The proposed attacks were tested on deep learning models using: (i) character-level, (ii) word-level, and (iii) hybrid (a combination of both). The experimental results indicated that the hybrid embedding classifier detected adversarial attacks with the best performance accuracy.

Instead of employing CNN as a single classifier, Rasymas & Dovydaitis [10] proposed an ensemble architecture integrating CNN with GRU for better detection accuracy. The proposed model also combined character-level and word-level embedding to extract features from the website URLs. These features were extracted automatically using deep learning algorithms and combined with 60 artificial features to improve the detection accuracy. The authors conducted several experiments by using different combinations of features from character embedding, word embedding, and lexical (artificial). The obtained results revealed that lexical features had no significant impact on the model's performance. Character-level and word-level embedding features, on the other hand, achieved the best detection accuracy of 94.4%.

In a similar approach, Chai et al. [11] made use of different deep learning algorithms to extract multiple features from the webpage content. Specifically, CNN was used to extract features from the visual design of the websites, while LSTM was utilized to obtain features from webpage text and URLs. Character and word embeddings were applied to features extracted using the LSTM classifier and further enhanced with an attention-based mechanism.

To sum up, character level and word level when used as stand-alone embedding techniques still have certain problems that need to be addressed. Although their hybrid approach offers better results, there still exists a lack of comparative analysis to benchmark different deep learning algorithms for detecting malicious URLs. This comparative study is crucial to provide a comprehensive examination of various deep learning techniques when character-level and word-level embedding methods are combined.

Table 1 provides a comparison between our paper and the previous studies using the hybrid text embedding method. Rasheed et al. [9] examined the adversarial attacks using three different types of text embedding: character-level, word-level, and hybrid. However, the authors used only CNN as a classifier and lack a comparative analysis with other deep learning techniques. Rasymas & Dovydaitis [10] enhanced the traditional CNN algorithm by combining it with GRU in an ensemble model to improve the detection accuracy. Nevertheless, the use of artificial features still requires manual

feature engineering and human intervention. Chai et al. [11] further enhanced the ensemble architecture by applying the attention-based mechanism. Yet, the authors used multiple features from the webpage content (visual, textual) and URLs. Extraction of these multi-dimensional features causes computation complexity that needs to be avoided. Unlike any of the previous studies, this paper only used automatic features to prevent manual feature engineering and third-party service dependency. In addition, a comparative analysis was also conducted to examine the state-of-the-art deep learning technologies in detecting malicious URLs.

Table 1. Comparison with the previous studies using hybrid text embedding method.

Reference	Features	Algorithm	Remark
[9]	Automatic	CNN	Lack a comparative analysis with other deep learning techniques
[10]	Automatic + Artificial	CNN-GRU	Require manual feature engineering due to artificial features
[11]	Automatic	CNN-LSTM	Computational complexity due to the extraction of multiple features
Our study	Automatic	CNN, BiGRU, BiLSTM	Provide a comprehensive analysis of state-of-the-art technologies

3. The Proposed Model

Motivated by the above, this paper provides a comparative study on several phishing detection models based on distributed representation and deep learning to solve the limitations of the existing literature. Specifically, the proposed models in this paper first extract features automatically from website URLs, then combine character-level and word-level embedding methods to obtain the vector representations, and finally apply three different deep learning techniques (CNN, BiGRU, and BiLSTM) to distinguish between legitimate and phishing URLs. Among deep learning techniques, CNN is one of the most popular algorithms due to its ability to extract features effectively and efficiently. GRU and LSTM are two variants of Recurrent Neural Network (RNN) that overcome the limitations of the conventional RNN model. However, GRU and LSTM employ unidirectional architecture which only considers the forward direction and ignores the backward direction of information. As a result, BiGRU and BiLSTM were used in our study to investigate the most recent advancement of deep learning techniques in the phishing detection domain.

The proposed models consist of two tiers: the first tier used character-level embedding, while the second tier applied word-level embedding. The outputs from these two tiers were concatenated and fed into deep learning models for feature extraction and classification. Both tiers took the URL strings as inputs and produced a single output to predict if the URL is malicious or benign. The architecture of the malicious URL detection model is demonstrated in Figure 1. The proposed model is comprised of five stages: 1) Webpage parsing, 2) Data pre-processing, 3) Feature embedding, 4) Feature extraction, and 5) Classification.

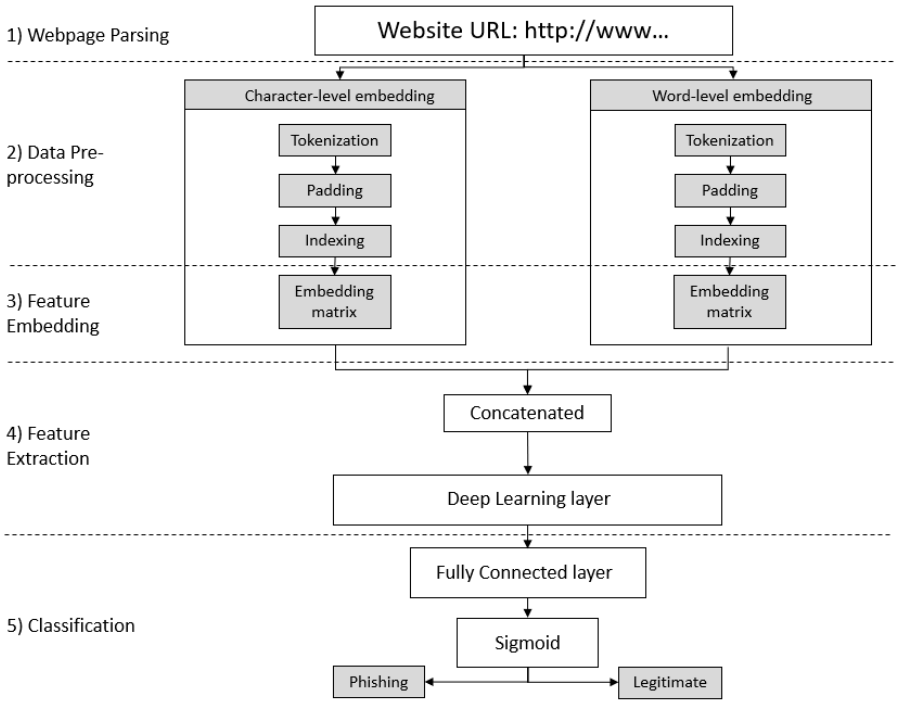


Figure 1. The proposed model architecture.

3.1. Webpage Parsing

The first stage in the implementation of the proposed model is webpage parsing. In this stage, the website prefixes (e.g., http, https, www) are removed from the original URLs since they do not have any significant impact on the classification decision. After deleting these prefixes, the remaining parts of the URLs and their respective classes (malicious or benign) are organized in a matrix that can be passed to the next stage for pre-processing.

3.2. Data Pre-processing

The second stage is data pre-processing which involves several steps, such as tokenization, encoding, and padding.

Tokenization is a mandatory process in any NLP task that is needed to split the input text data into smaller parts before performing any further pre-processing steps. This is because website URLs are typically in a string format which can be tokenized into a sequence of characters or words.

The next step in the data pre-processing stage is to encode the sequence of tokens in numerical form. For example, in character-level embedding, each character in the original URL is encoded and replaced by a unique number. In word-level embedding, each word is separated by special characters, such as ".", "@", "/", etc., and assigned with a numerical value to establish a word-level corpus of the URL.

After constructing the character-level and word-level corpora, each tokenized sequence is normalized to a fixed-size length. If the URL length is longer than the maximum length, the additional characters/words are truncated. Otherwise, the missing characters/words are replaced by the <PAD> tags at the end of the URL.

3.3. Feature Embedding

In the feature embedding stage, the original URL is mapped into a low-dimensional vector that can be given as an input to the subsequent deep learning models. During this process, the encoded URL is converted to a two-dimensional dense matrix that can be recognized by the deep neural network. Initially, the website URL is transformed into a one-dimensional matrix. Then, the embedding matrix is obtained by transforming the matrix into a two-dimensional dense matrix that can be processed by deep learning algorithms.

3.4. Feature Extraction

The embedding matrices from character-level and word-level embeddings are concatenated and fed into three deep learning classifiers, including CNN, BiGRU, and BiLSTM. CNN structure contains three basic layers: convolutional, pooling, and fully connected layer. The first layer is used to perform the convolution operation and the second layer is utilized to reduce the feature dimension generated due to the convolution operation.

Between the two variants of RNN, GRU is a lightweight algorithm and has similar or even better performance when training on small datasets compared to LSTM. Both variants have processing units with memory functions that can handle time-series or sequential data. Moreover, BiGRU and BiLSTM consider both forward and backward directions of information.

3.5. Classification

Lastly, the outcome from a deep learning classifier is passed through a fully connected layer before using the sigmoid function to determine if the URL is malicious or benign. Table 2 provides the parameter settings for text embedding and deep learning used in this study.

Table 2. Parameter settings for text embedding and deep learning

	CNN	**BiGRU**	**BiLSTM**
Max URL length	500	500	500
Embedding size	64	64	64
Dropout rate	0.5	0.5	0.5
Learning rate	0.005	0.005	0.005
Batch size	512	512	512
Epoch	10	10	10
Filter size	3	-	-
Number of units	32	16	16

4. Experiment and Results

In this paper, the distributed representation methods and deep learning techniques were implemented using Python 3.6 programing language and Tensorflow 2.7.0, on Windows 10 64-bit operating system and Google Colaboratory (11.441MB CPU capacity and 12.8 GB CPU RAM). Ebbu2017 Phishing Dataset [12] was used to train the proposed deep learning models. It is a public dataset published by Sahingoz et al. [12] which is available on GitHub [13]. The dataset contains a total of 73,575 URLs, out of which 37,175 URLs are malicious and 36,400 are benign URLs.

The results obtained from the experiments are displayed in Table 3 and Figure 2. It is observed that character-level and word-level when combined with deep learning technologies can significantly improve the performance of phishing detection models. In addition, among three deep learning classifiers, CNN produced the best performance across all calculated metrics. It had the lowest measures in FPR, FNR, training time, and testing time while achieving the highest measures in the other performance metrics. Compared to the CNN algorithm, the training and testing time of two RNN variants were significantly longer. Meanwhile, the performance of BiGRU and BiLSTM were almost similar to each other, with the former being slightly better than the latter.

Table 3. Evaluation metrics of three deep learning classifiers.

Classifier	FPR (%)	FNR (%)	Pr (%)	Rc (%)	F1 (%)	Acc (%)	Training time (min)	Testing time (s)
CNN	1.17	0.64	99.36	98.84	99.10	99.10	18.41	10.80
BiGRU	1.35	1.09	98.91	98.67	98.79	98.78	35.45	33.00
BiLSTM	1.95	1.48	98.52	98.10	98.31	98.29	36.46	41.61

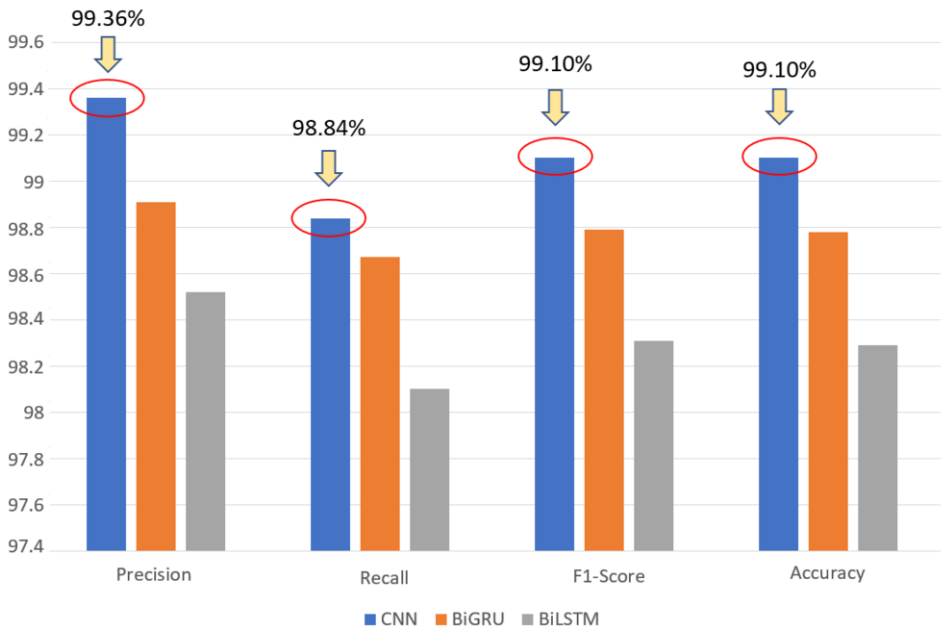


Figure 2. Performance metrics of three deep learning models.

Figure 3 illustrates the Receiver Operating Characteristic (ROC) curves of three deep learning algorithms. Area Under the Curve (AUC) is the total area measured under the ROC curve, which represents the model's ability in classifying malicious URLs. The closer the AUC to 1, the better the phishing detection model. As the AUC values of three deep learning classifiers were slightly different from each other, their ROC curves were almost overlapped. This indicates all three models are comparable in detecting phishing websites and they can be fused to form an ensemble deep learning architecture to further improve the performance accuracy.

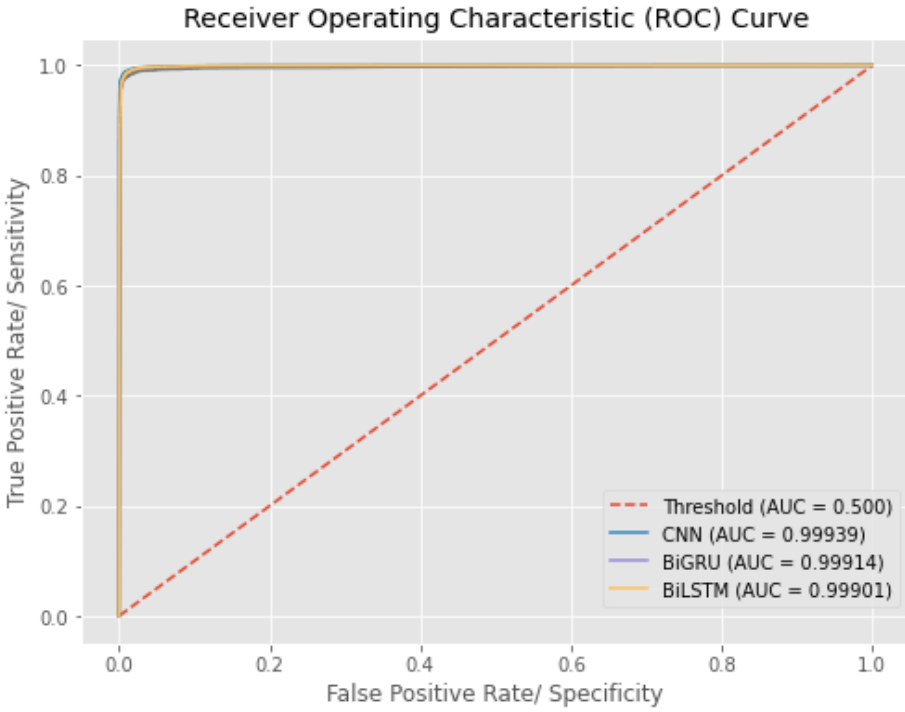


Figure 3. Performance metrics of three deep learning models.

Furthermore, to investigate the effects of various text embedding methods on phishing detection accuracy, we carried out another set of experiments. Since CNN and BiGRU produced better performance than BiLSTM, they were used as baseline classifiers while different text embedding techniques were examined. Table 4 and Table 5 summarize the experimental results obtained from applying character-level, word-level, and hybrid (character and word) embedding approaches to these two classifiers. The findings revealed that the hybrid approach outperformed the individual methods in classifying malicious URLs as it achieved the highest detection accuracy regardless of the deep learning algorithm. Moreover, word and hybrid embedding methods are significantly better than character-level embedding alone. This is partly because character embedding fails to capture the semantic and syntactic meaning between

characters in the input sequence, while word embedding can discover the relationship among words in the data string.

Table 4. Effect of text embedding methods on detection accuracy using CNN classifier.

Classifier	FPR (%)	FNR (%)	Pr (%)	Rc (%)	F1 (%)	Acc (%)
Character CNN	2.20	3.80	96.20	97.84	97.01	96.98
Word CNN	1.70	0.69	99.31	98.32	98.81	98.80
Hybrid CNN	1.17	0.64	99.36	98.84	99.10	**99.10**

Table 5. Effect of text embedding methods on detection accuracy using BiGRU classifier.

Classifier	FPR (%)	FNR (%)	Pr (%)	Rc (%)	F1 (%)	Acc (%)
Character BiGRU	6.01	2.19	97.81	94.09	95.92	95.88
Word BiGRU	6.58	2.08	97.92	99.34	98.63	98.63
Hybrid BiGRU	1.35	1.09	98.91	98.67	98.79	**98.78**

5. Conclusion and Future Works

In this paper, we conducted a comparative analysis of three different phishing detection models using distributed representation and deep learning technologies. To overcome the limitations of the existing malicious classification solutions, we combined character-level and word-level embedding methods to obtain the vector representations from the website URLs. Moreover, we also employed three deep learning algorithms, including CNN, BiGRU, and BiLSTM, for classification. Our experimental results indicated that CNN was the best model since it outperformed the other two models in both detection accuracy and computational time. Examination of different embedding methods also revealed that the fusion of character and word embedding techniques produced higher accuracy than individual approaches. For future research, we plan to explore other text representation methods, such as Word2vec, Glove Vector for word representation (GloVe), or FastText. We are also going to implement ensemble models by integrating different deep learning algorithms for better accuracy.

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